

Question 1

Objective

To obtain the building plans or dimensions of buildings of interest that can be used for testing.

Attempts

We have fitted the following building footprints to 1000 Burnaby lots:

- 20x25 ft
- 15x20 ft
- 10x15 ft

Evidence

- The building footprints along with the Python scripts used to generate them are here:
 - o [20x25 ft](#)
 - o [15x20 ft](#)
 - o [10x15 ft](#)

We will write a detailed report soon, but briefly, we use the same algorithm used to label lot boundaries to instead label the footprint boundaries. Then, setbacks are applied to footprint boundaries based on labels.

- The current result for 1000 Burnaby lots can be found here:
 - o [Directory in git](#)
 - o [Script to generate buildable areas without considering setbacks from existing structures](#)
 - o [Script to fit footprint on buildable areas](#)

We will write a detailed report soon, but briefly, the footprint is moved around in steps within each parcel. The average bearing of the front footprint boundaries must match the average bearing of the rear lot boundaries within a certain tolerance. The footprint must be closer to the rear than the front for placement to be valid.



Diagnosis

We don't know what building dimensions are standard for housing developers interested in building laneway homes.

Proposed Next Steps

Provide us with the building plans or dimensions of any buildings of interest you may have.

Decision Ask

Do you have any buildings of interest with plans or dimensions that can be shared so we can avoid hardcoding arbitrary building dimensions?

Question 2

Objective

To determine when different types of setbacks should be applied to front, side, and rear lot boundaries.

Attempts

The currently we assume that if no parcels touch a side lot boundary, then the **Street Flanking** setback should be used. However, if it is a lane on that side, the **Interior Side** setback should be used instead.

Also, we assume that if no parcels touch a rear lot boundary, then the **Lane** setback should be used. However, if it is a street on that side, the **Street Flanking** setback should be used instead.

Evidence

Here is a report of the current algorithm used to apply setbacks based on lot boundary labels (front, side, rear). Focus on the **Applying Setbacks to Define Buildable Area** section starting on page 3:

https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/blob/main/code/gis_files/reports/buildable_area_algorithms.pdf

Diagnosis

The errors in the algorithm were not obvious until implemented in QGIS.

Proposed Next Steps

The purposed changes are outlined in the table below in **red**.

Current Algorithm	Recommended Algorithm
1) If the boundary line is FRONT, apply Street Non-Flanking setback 2) If the boundary line is SIDE: a) If the boundary line touches another parcel (endpoints excluded), apply Internal Side setback b) Else, apply Street Flanking setback 3) If the boundary line is REAR: a) If the boundary line touches another parcel (endpoints excluded), apply Internal Rear setback b) Else, apply Lane setback	1. If the boundary line is FRONT, apply Street Non-Flanking setback 2. If the boundary line is SIDE: a. If the boundary line touches another parcel (endpoints excluded), apply Internal Side setback b. Else: i. If the boundary line's midpoint is closer to a street than a lane, apply Street Flanking setback ii. Else, apply Internal Side setback 3. If the boundary line is REAR: a. If the boundary line touches another parcel (endpoints excluded), apply Internal Rear setback b. Else: i. If the boundary line's midpoint is closer to a street than a lane, apply Street Flanking setback ii. Else, apply Lane setback

Decision Ask

Do you see any edge cases that the recommended algorithm does not capture when applying setbacks based on lot boundary labels (front, side, rear).